



L OVELY
P ROFESSIONAL
U NIVERSITY

Transforming Education Transforming India

Project Title:

Cross-Country Analysis of Economic Growth
Trends Using GDP (Current US\$), 1970–2024

Course: CSD 301 - Data Analytics Project

Group Members: Rasleen Kaur (08), Sparsh Verma (05)

1 Introduction

◆ Problem Statement

To perform a descriptive analysis of country-wise GDP (Current US\$) from 1970 to 2024 by examining long-term growth trends, calculating growth rates, analyzing economic fluctuations, and comparing relative economic progression among nations.

◆ Project Objective

- Study long-term GDP trends across countries
- Measure economic growth using CAGR
- Analyze growth volatility
- Compare economic performance over time
- Identify expansion patterns among nations

Dataset Selection Journey

Before finalizing the GDP dataset for this project, two alternative datasets were initially considered. However, after preliminary inspection and feasibility analysis, both were rejected due to structural and analytical limitations.

1 Dataset 1: Depression and Anxiety Dataset

Initial Objective

The dataset was intended for analyzing mental health patterns related to depression and anxiety indicators.

Issues Identified

- The dataset contained primarily categorical and textual variables.
- Only one column contained meaningful numerical values suitable for quantitative analysis.
- Lack of multiple numeric features restricted statistical modeling, correlation analysis, and feature engineering.
- Limited scope for meaningful descriptive analytics or multivariate exploration.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Indicator	Group	State	Subgroup	Phase	Time Period	Time Period L	Time Period St	Time Period E	Value	Low CI	High CI	Confidence Int	Quartile Range
2	Symptoms of f	National Estin	United States	United States	1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			23.5	22.7	24.3	22.7 - 24.3	
3	Symptoms of f By Age	United States	18 - 29 years		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			32.7	30.2	35.2	30.2 - 35.2	
4	Symptoms of f By Age	United States	30 - 39 years		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			25.7	24.1	27.3	24.1 - 27.3	
5	Symptoms of f By Age	United States	40 - 49 years		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			24.8	23.3	26.2	23.3 - 26.2	
6	Symptoms of f By Age	United States	50 - 59 years		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			23.2	21.5	25	21.5 - 25.0	
7	Symptoms of f By Age	United States	60 - 69 years		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			18.4	17	19.7	17.0 - 19.7	
8	Symptoms of f By Age	United States	70 - 79 years		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			13.6	11.8	15.5	11.8 - 15.5	
9	Symptoms of f By Age	United States	80 years and a		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			14.4	9	21.4	9.0 - 21.4	
10	Symptoms of f By Sex	United States	Male		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			20.8	19.6	22	19.6 - 22.0	
11	Symptoms of f By Sex	United States	Female		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			26.1	25.2	27.1	25.2 - 27.1	
12	Symptoms of f By Race/Hispa	United States	Hispanic or La		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			29.4	26.8	32.1	26.8 - 32.1	
13	Symptoms of f By Race/Hispa	United States	Non-Hispanic		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			21.4	20.6	22.1	20.6 - 22.1	
14	Symptoms of f By Race/Hispa	United States	Non-Hispanic		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			25.6	23.7	27.5	23.7 - 27.5	
15	Symptoms of f By Race/Hispa	United States	Non-Hispanic		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			23.6	20.3	27.1	20.3 - 27.1	
16	Symptoms of f By Race/Hispa	United States	Non-Hispanic		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			28.3	24.8	32	24.8 - 32.0	
17	Symptoms of f By Education	United States	Less than a hi		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			32.7	27.8	38	27.8 - 38.0	
18	Symptoms of f By Education	United States	High school di		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			25.4	23.9	26.9	23.9 - 26.9	
19	Symptoms of f By Education	United States	Some college/		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			25.6	24.4	26.9	24.4 - 26.9	
20	Symptoms of f By Education	United States	Bachelor's deg		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			17.6	16.8	18.4	16.8 - 18.4	
21	Symptoms of f By State	Alabama	Alabama		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			18.6	14.6	23.1	14.6 - 23.1	16.5 - 20.7
22	Symptoms of f By State	Alaska	Alaska		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			19.2	16.8	21.8	16.8 - 21.8	16.5 - 20.7
23	Symptoms of f By State	Arizona	Arizona		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			22.4	19.4	25.5	19.4 - 25.5	22.2 - 24.0
24	Symptoms of f By State	Arkansas	Arkansas		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			26.6	22.3	31.3	22.3 - 31.3	24.1 - 28.7
25	Symptoms of f By State	California	California		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			25.4	22.5	28.6	22.5 - 28.6	24.1 - 28.7
26	Symptoms of f By State	Colorado	Colorado		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			22	19.4	24.9	19.4 - 24.9	20.8 - 22.1
27	Symptoms of f By State	Connecticut	Connecticut		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			24.4	20.1	29.1	20.1 - 29.1	24.1 - 28.7
28	Symptoms of f By State	Delaware	Delaware		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			21.1	17.6	24.9	17.6 - 24.9	20.8 - 22.1
29	Symptoms of f By State	District of Col	District of Col		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			26.4	22.1	31.1	22.1 - 31.1	24.1 - 28.7
30	Symptoms of f By State	Florida	Florida		1	1 Apr 23 - May 5, 04/23/2020	05-05-2020			22.5	19.7	25.4	19.7 - 25.4	22.2 - 24.0

Reason for Rejection

Since the project requires structured numerical time-series or multivariate data for analytical depth, the dataset was deemed unsuitable. The absence of sufficient numeric variables made it difficult to perform meaningful growth, trend, or comparative analysis.

2 Dataset 2: MACRO-LEVEL ECONOMIC Analysis of Different States Of India

Initial Objective

This dataset was explored for macro-level economic or statistical analysis.

Issues Identified

- The dataset structure was highly clustered and aggregated.
- Data appeared grouped in consolidated formats rather than clean country-wise or row-wise structure.
- Significant preprocessing would be required to restructure the dataset.
- Inconsistent formatting and possible hierarchical grouping reduced usability.
- Limited flexibility for time-series or cross-sectional analysis.
- Data complexity could overshadow analytical clarity within CA-1 scope.

2 Project Vision / Approach

◆ Vision

To understand how economies evolved over five decades and assess relative economic progression using structured data analysis.

◆ Methodology

1. Dataset inspection and validation
2. Filtering the GDP indicator
3. Handling missing values
4. Restricting analysis period (1980–2024)
5. Feature engineering:
 - Growth Functions
 - CAGR
 - GDP Multiplier
 - Growth

◆ Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook
- Visual Studio Code
- Git, GitHub

GitHub Repository Link: <https://github.com/Sparshcoder18/gdp-growth-analysis>

3 Dataset Details

◆ Dataset Source

United Nations Statistics Division (UN Data Portal)

Dataset: <https://unstats.un.org/unsd/amaapi/api/file/2>

◆ Type of Data

- Time-series macroeconomic data
- Numerical (float values for GDP)
- Cross-country panel dataset

◆ Dataset Size

- Initial shape: 3715 rows × 58 columns
- After GDP filtering: 220 rows × 58 columns
- Final cleaned dataset: (mention your final shape from df_clean)

◆ Important Features

- Country
- Year columns (1970–2024)
- GDP values

4 Preprocessing & Feature Engineering

◆ Data Cleaning Steps

- Removed irrelevant rows
- Filtered only the GDP indicator
- Dropped unnecessary columns
- Checked duplicate countries
- Analyzed missing values
- Removed countries with excessive missing data
- Restricted analysis to 1980–2024

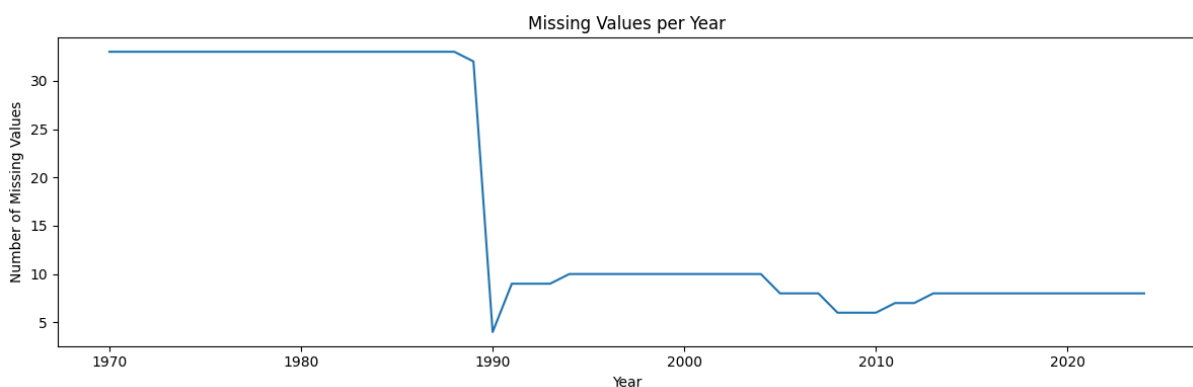


Figure 1: Missing Values per Year (1970–2024)]

A year-wise missing value analysis was conducted to assess data completeness. The visualization above shows a sharp decline in missing values after the early 1990s.

◆ Preprocessing Techniques

- Missing value analysis (country-wise & year-wise)
- Data type validation
- Index reset
- Column filtering

◆ Feature Engineering

- CAGR (1980–2024)
- GDP Multiplier
- Growth calculation functions

Include formula:

$$CAGR = \left(\frac{GDP_{end}}{GDP_{start}} \right)^{1/n} - 1$$



Detailed Description of Dataset Columns

◆ Original Dataset Columns

After filtering the dataset to retain only the GDP (Current US\$) indicator, the dataset contains the following categories of columns:

1 Country

Type: Object (Categorical)

Description: Represents the name of each country included in the dataset. Each row corresponds to one country.

Purpose in Analysis: Used as the primary identifier for country-wise comparison of GDP trends and growth metrics.

2 Year Columns (1970–2024)

Type: Float64 (Numerical)

Description: Each column represents GDP (Current US\$) for a specific year.

Example:

- 1970 → GDP value for the year 1970
- 1980 → GDP value for the year 1980
- 2024 → GDP value for the year 2024

Data Characteristics:

- Time-series format (wide structure)
- Values expressed in Current US Dollars
- Some missing values in earlier years

Purpose in Analysis:

- Trend analysis over time
 - Growth rate computation
 - CAGR calculation
 - Economic expansion comparison
-

Helper Columns (Temporary Columns Used During Cleaning)

These were created during preprocessing but removed in the final dataset.

③ Missing_Count

Type: Integer

Description: Total number of missing GDP values per country across all years.

Purpose: To identify countries with excessive missing data.

④ Missing_1980_2024

Type: Integer

Description: Total missing values within the defined analysis period (1980–2024).

Purpose: To filter countries and ensure the reliability of long-term growth analysis.

Feature Engineered Columns

After preprocessing, the following analytical features were created:

⑤ CAGR (1980–2024)

Type: Float64 (Percentage)

Formula:

$$CAGR = \left(\frac{GDP_{2024}}{GDP_{1980}} \right)^{1/n} - 1$$

Where: $n=44$

Description: Represents the compound annual growth rate of GDP between 1980 and 2024.

Interpretation:

- Higher CAGR → Faster long-term economic growth
- Lower CAGR → Slower economic progression

Purpose: Used to compare sustained long-term economic growth across countries.

6 GDP_Multiplier

Type: Float64

Formula:

$$GDP\ Multiplier = \frac{GDP_{2024}}{GDP_{1980}}$$

Description: Shows how many times a country's GDP increased between 1980 and 2024.

Interpretation:

- Value = 2 → Economy doubled
- Value = 5 → Economy expanded fivefold

Purpose: Used to measure total economic expansion over the analysis period.

7 Growth_Volatility

Type: Float64

Description: Standard deviation of annual GDP growth rates between 1980–2024.

Interpretation:

- High value → Economically unstable growth

- Low value → Stable growth pattern

Purpose: Used to analyze economic stability and fluctuation intensity.

Functions Implemented in the Project

To avoid the unnecessary creation of multiple growth columns, reusable functions were implemented.

① `compute_growth(df, year)`

Purpose: Calculates GDP growth (%) for a selected year compared to the previous year.

Formula:

$$Growth_t = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \times 100$$

Use Case: To analyze short-term annual growth.

② `compute_growth_between(df, start_year, end_year)`

Purpose: Calculates the total percentage GDP growth between two selected years.

Formula:

$$Growth = \frac{GDP_{end} - GDP_{start}}{GDP_{start}} \times 100$$

Use Case: To compare economic growth across selected periods.

③ `compute_cagr(df, start_year, end_year)`

Purpose: Calculates compound annual growth rate between two selected years.

Why Used Instead of Simple Average Growth?

- Removes yearly volatility distortion
 - Provides a smoothed long-term growth rate
 - More accurate for macroeconomic comparison
-

Final Dataset Structure

After cleaning and feature engineering:

- Rows → Countries (after filtering excessive missing values)
- Columns →
 - Country
 - Year columns (1980–2024 used for analysis)
 - GDP_Multiplier
 - Growth_Volatility

This structured dataset enables reliable descriptive cross-country economic comparison.



5 Key Findings

1 Missing Data Pattern Justifies Analysis Period

The missing values visualization shows a high concentration of missing data prior to the early 1990s, followed by a significant improvement in data completeness. This supports the decision to restrict detailed growth analysis to the 1980–2024 period and avoid forward-filling, thereby maintaining analytical integrity.

2 Strong Divergence in Long-Term GDP Trends

The GDP trend comparison between India, the United States, and China highlights substantial divergence in economic trajectories:

- The United States maintained steady and dominant economic growth throughout the period.
- China exhibited accelerated growth, particularly after the early 2000s, indicating structural economic transformation.
- India demonstrated consistent upward growth, though at a comparatively lower absolute GDP scale.

This reflects differing development stages and growth strategies among major economies.

**To compare long-term economic trajectories,
the GDP trends of major economies were visualized.**

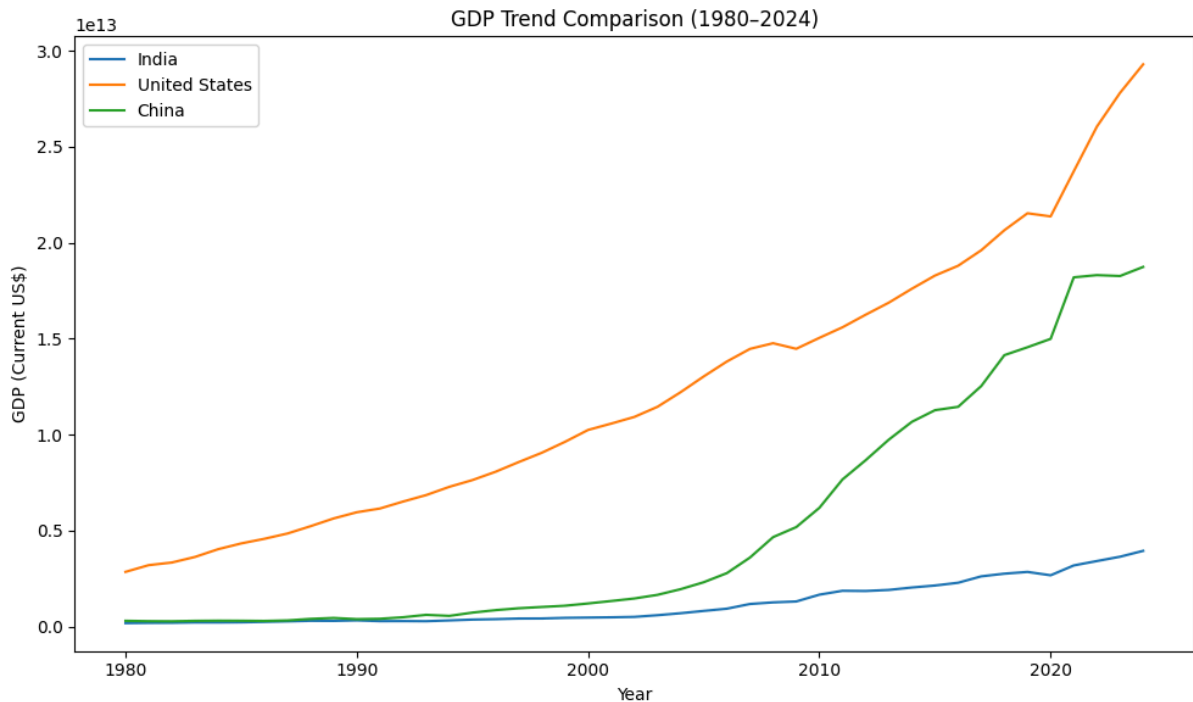


Figure 2: GDP Trend Comparison (1980–2024)

From 1980–2024, the United States maintained the highest GDP, China showed the fastest growth (especially after 2000), and India grew steadily but remained far behind both.

3 Presence of High Growth Volatility in Certain Economies

The Growth Volatility ranking reveals that several countries, including Equatorial Guinea, Iraq, Angola, and Libya, experienced high fluctuations in GDP growth. This indicates economic instability, possibly influenced by:

- Resource dependence
- Political instability
- External economic shocks

High volatility suggests that strong growth in some periods may have been accompanied by sharp contractions in others.

Growth volatility was analyzed to assess economic stability across countries.

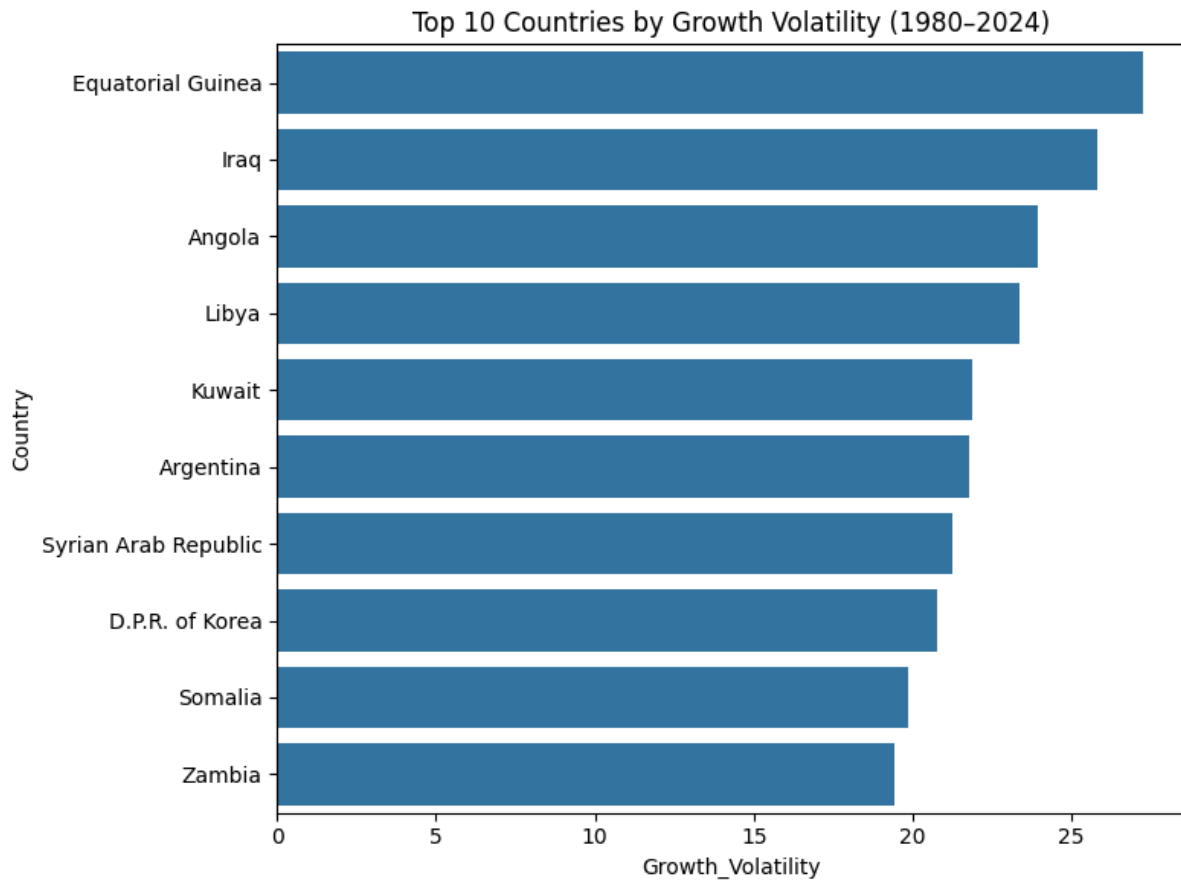


Figure 3: Top 10 Countries by Growth Volatility (1980–2024)

Equatorial Guinea shows the highest GDP growth volatility, followed by Iraq and Angola, indicating these economies experienced the most unstable and fluctuating growth between 1980–2024.

4 Stability vs. Growth Trade-Off

The analysis suggests a contrast between:

- Economies with stable, moderate growth (typically developed nations)
- Economies with high but volatile growth patterns (often developing or resource-dependent nations)

This demonstrates that high long-term growth does not necessarily imply economic stability.

5 Multi-Dimensional Economic Assessment

By combining CAGR, GDP Multiplier, and Growth Volatility, the study provides a comprehensive evaluation of economic performance across countries, covering:

- Growth rate
- Scale of expansion
- Stability of growth

This multi-metric approach strengthens cross-country comparison.

6 Conclusion

This study conducted a descriptive cross-country analysis of GDP (Current US\$) spanning 1970–2024, with a focused analysis on the 1980–2024 period to ensure data reliability. Through structured preprocessing and feature engineering, key growth metrics such as CAGR, GDP Multiplier, and Growth Volatility were derived.

The findings reveal significant disparities in long-term economic progression across nations. Developed economies demonstrate stable and sustained growth patterns, while several developing and resource-dependent countries exhibit higher volatility. Emerging economies such as China have shown accelerated structural expansion over the past two decades.

The integration of growth rate, expansion magnitude, and stability metrics provides a holistic understanding of global economic evolution. Overall, the analysis highlights the diversity in

economic trajectories and emphasizes the importance of evaluating both growth intensity and stability when assessing long-term economic performance.